OPERATING SYSTEMS

Kernel Data Structures and Computing Environments

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Slides Credits for all the PPTs of this course

- The slides/diagrams in this course are an **adaptation**, **combination**, and **enhancement** of material from the following resources and persons:

1. Slides of Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne - 9th edition 2013 and some slides from 10th edition 2018
2. Some conceptual text and diagram from Operating Systems - Internals and Design Principles, William Stallings, 9th edition 2018
3. Some presentation transcripts from A. Frank – P. Weisberg
4. Some conceptual text from Operating Systems: Three Easy Pieces, Remzi Arpaci

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Kernel Data Structures

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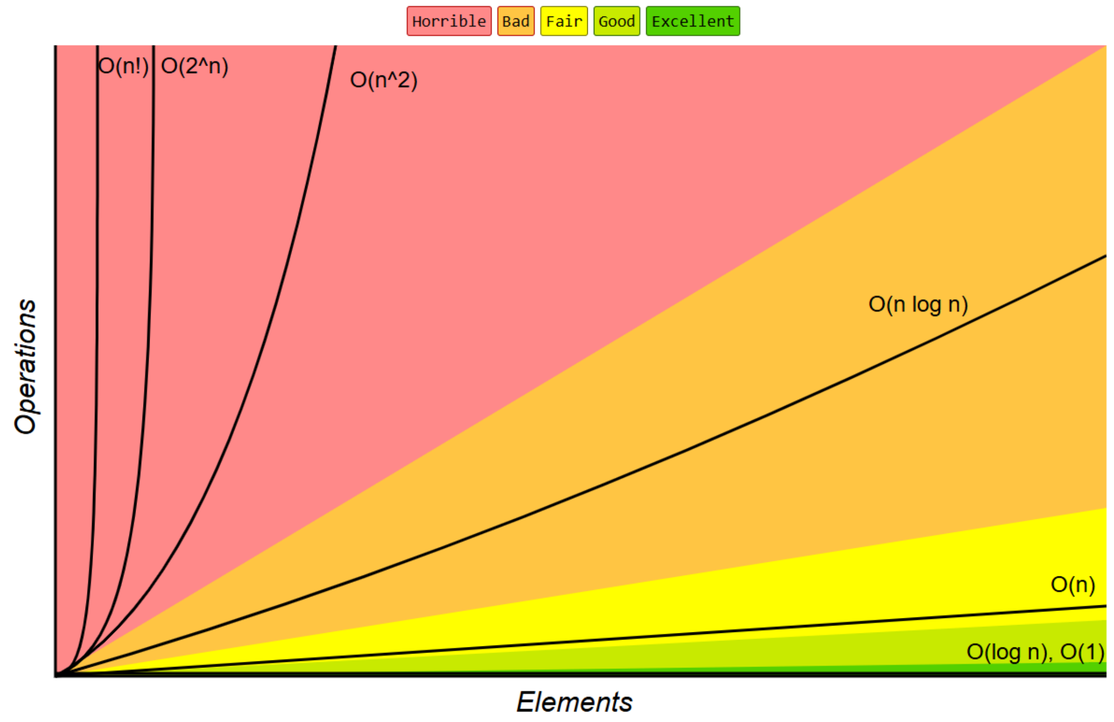
Kernel Data Structures



Common Data Structure Operations

Data Structure	Time Complexity							
	Average				Worst			
	Access	Search	Insertion	Deletion	Access	Search	Insertion	Deletion
Array	Θ(1)	Θ(n)	Θ(n)	Θ(n)	O(1)	O(n)	O(n)	O(n)
Stack	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)
Queue	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)
Singly-Linked List	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)
Doubly-Linked List	Θ(n)	Θ(n)	Θ(1)	Θ(1)	O(n)	O(n)	O(1)	O(1)
Skip List	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)
Hash Table	N/A	Θ(1)	Θ(1)	Θ(1)	N/A	O(n)	O(n)	O(n)
Binary Search Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)
Cartesian Tree	N/A	Θ(log(n))	Θ(log(n))	Θ(log(n))	N/A	O(n)	O(n)	O(n)
B-Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))
Red-Black Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))
Splay Tree	N/A	Θ(log(n))	Θ(log(n))	Θ(log(n))	N/A	O(log(n))	O(log(n))	O(log(n))
AVL Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(log(n))	O(log(n))	O(log(n))	O(log(n))
KD Tree	Θ(log(n))	Θ(log(n))	Θ(log(n))	Θ(log(n))	O(n)	O(n)	O(n)	O(n)

Array:

- An array is a simple data structure in which each element can be accessed directly.
- Main Memory constructed with array.
- How the data is accessed?
- Items with multiple bytes are accessed as item number × item size
- But what about storing an item whose size may vary?
- what about removing an item if the relative positions of the

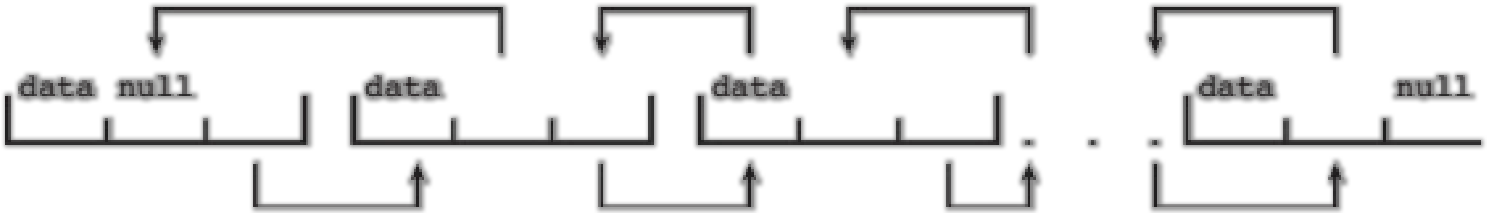
- Standard programming data structures are used extensively in OS

Singly linked list

- The items in a list must be accessed in a particular order.
- common method for implementing this structure is a linked list

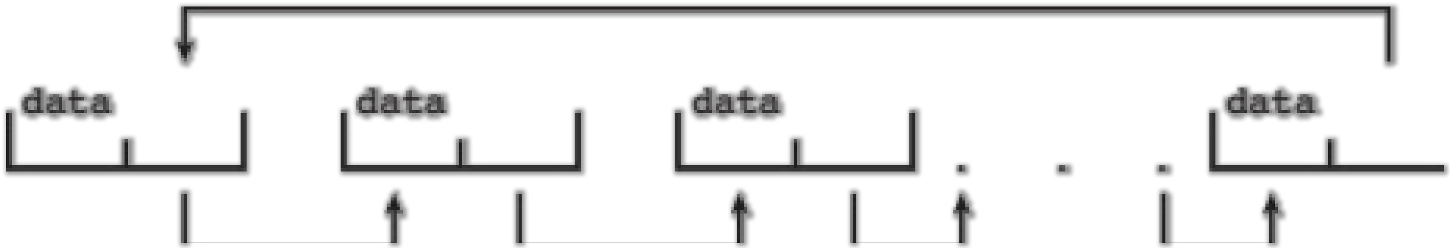
- In a **singly linked list,** each item points to its successor.

Doubly linked list



In a **doubly linked list**, a given item can refer either to its predecessor or to its successor.

Circular linked list



In a **circularly linked list**, the last element in the list refers to the first element, rather than to null.

Lists

Advantages:

- Linked lists accommodate items of varying sizes.
- Allow easy insertion and deletion of items

Disadvantages:

- Performance for retrieving a specified item in a list of size n is linear — O(n) , as it requires potentially traversing all n elements in the worst case.

Usage:

- Lists are used by the some of the kernel algorithms
- Constructing more powerful data structures such as stacks and

Stack - a sequentially ordered data structure that uses **LIFO** principle for adding and removing items

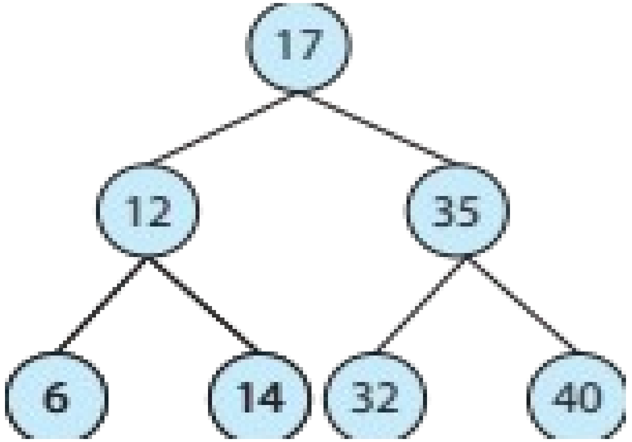
- OS often uses a stack when involving function calls.
- Parameters, local variables and the return address are **push**ed onto the stack when a function is called
- Return from the function call **pop**s those items off the stack

Queue - a sequentially ordered data structure that uses **FIFO** principle for adding and removing items

- Tasks waiting to be run on an available CPU are organized in queues

Trees

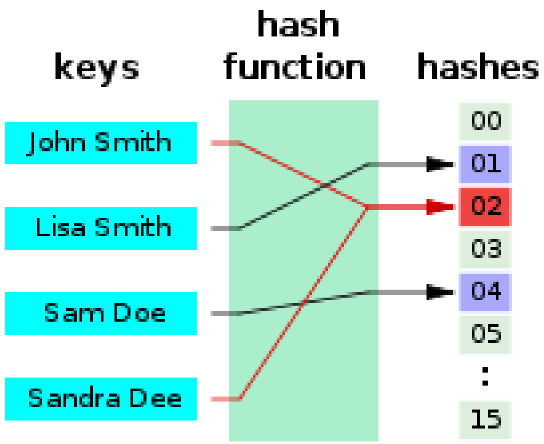
- Data structure used to represent data hierarchically.
- Data values in a tree structure are linked through parent–child relationships
- **Binary search tree**
  - ordering between 2 children: left <= right
  - Search performance is $O(n)$



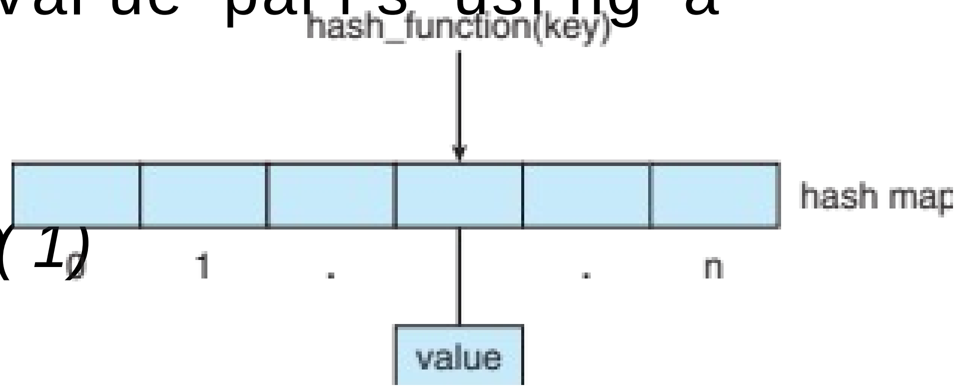
- **Balanced binary search tree –** (a tree containing n items has at most log n levels)
  - Search performance is $O(\log n)$

Hash Functions and Maps

- Hash functions can result in the same output value for 2 inputs
- **Hash function** can be used to implement a **hash map**
  - Maps



y: value pairs using a hash



  - Searc

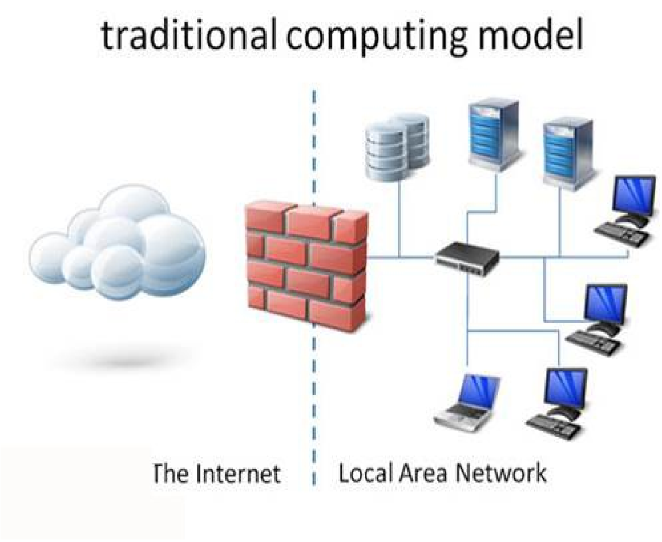
O(1)

Bitmap - string of *n* binary digits representing the status of *n* items

- Availability of each resource is indicated by the value of a binary digit
  - 0 – resource is available
  - 1 – resource is unavailable
- Value of the **i**th position in the bitmap is associated with the **i**th resource
  - Example: bitmap 001011101 shows resources 2, 4, 5, 6, and 8 are

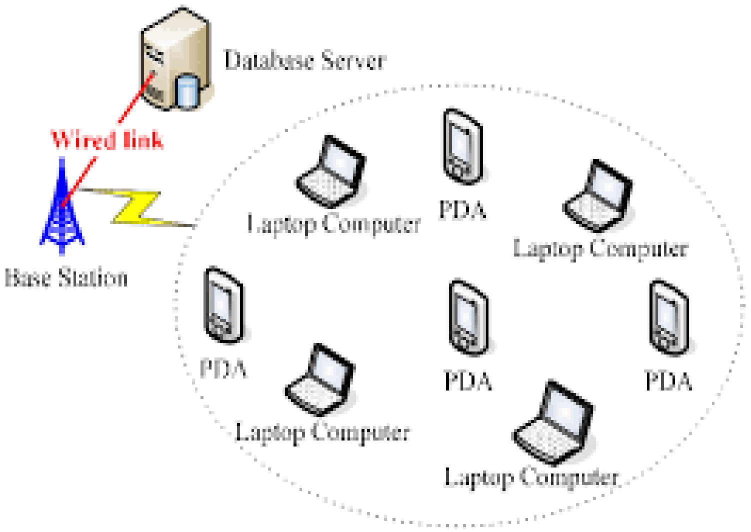
Computing Environments – Traditional

- Stand-alone general purpose machines
- But blurred as most systems interconnect with others (i.e., the Internet)
- **Portals** provide web access to internal systems
- **Network computers** (**thin clients**) are like Web terminals
- Mobile computers interconnect via **wireless networks**



Computing Environments – Mobile

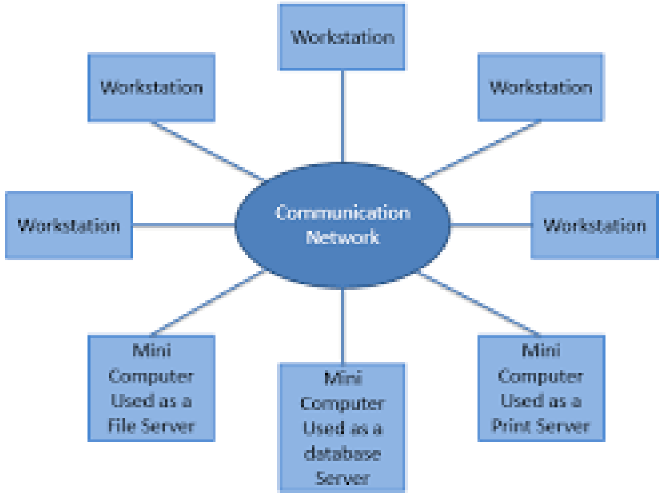
- Handheld smartphones, tablets, etc
- What is the functional difference between them and a "traditional" laptop?
- Extra feature – more OS features (GPS, gyroscope)
- Allows new types of apps like ***augmented reality***
- Use IEEE 802.11 wireless, or cellular data networks for connectivity



Computing Environments – Distributed

Distributed computing

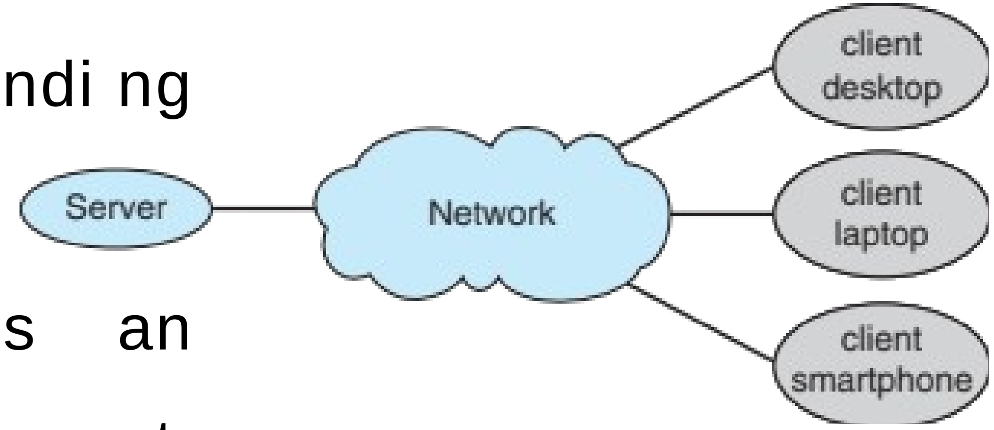
- Collection of separate, possibly heterogeneous systems networked together
    - **Network** is a communications path, **TCP/IP** most
        - **Local Area Network** (**LAN**)
        - **Wide Area Network** (**WAN**)
        - **Metropolitan Area Network** (**MAN**)
        - **Personal Area Network** (**PAN**)



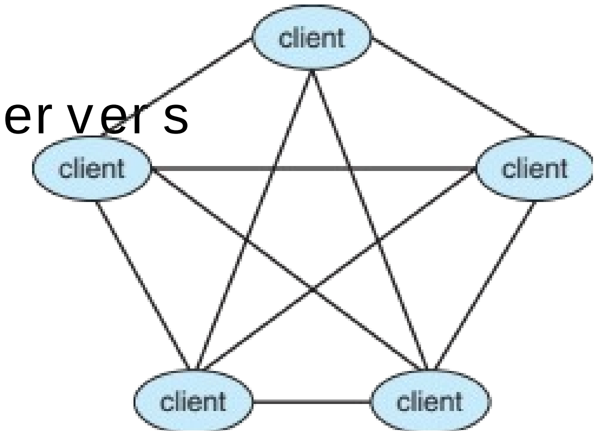
- **Network Operating System** provides features between systems across network
    - Communication scheme allows systems to exchange messages

Client-Server Computing

- Dumb terminals replaced by smart PCs
- Many systems now **servers**, responding to requests generated by **clients**
  - **Compute-server system** provides an interface to client to request services (i.e., database)
  - **File-server system** provides interface for clients to store and



Computing Environments – Peer-to-Peer



- Another model of distributed system
  - P2P does not distinguish clients and servers
  - Instead all nodes are considered peers
  - May each act as client, server or both
  - Node must join P2P network
    - Registers its service with central lookup service on network, or
    - Broadcast request for service and respond to requests for service via ***discovery protocol***
  - Examples include Napster and Gnutella, **Voice over IP** (**VoIP**) such as Skype

Computing Environments – Virtualization

- Allows operating systems to run applications within other OSes
  - Vast and growing industry
- **Emulation** used when source CPU type different from target type (i.e. PowerPC to Intel x86)
  - Generally slowest method
  - When computer language not compiled to native code – **Interpretation**
- **Virtualization** – OS natively compiled for CPU, running **guest** OSes also natively compiled
  - Consider VMware running WinXP guests, each running applications, all on native WinXP **host** OS

Computing Environments – Virtualization

- Use cases involve laptops and desktops running multiple OSes for exploration or compatibility
  - Apple laptop running Mac OS X host, Windows as a guest
  - Developing apps for multiple OSes without having multiple systems
  - QA testing applications without having multiple systems
  - Executing and managing compute environments within

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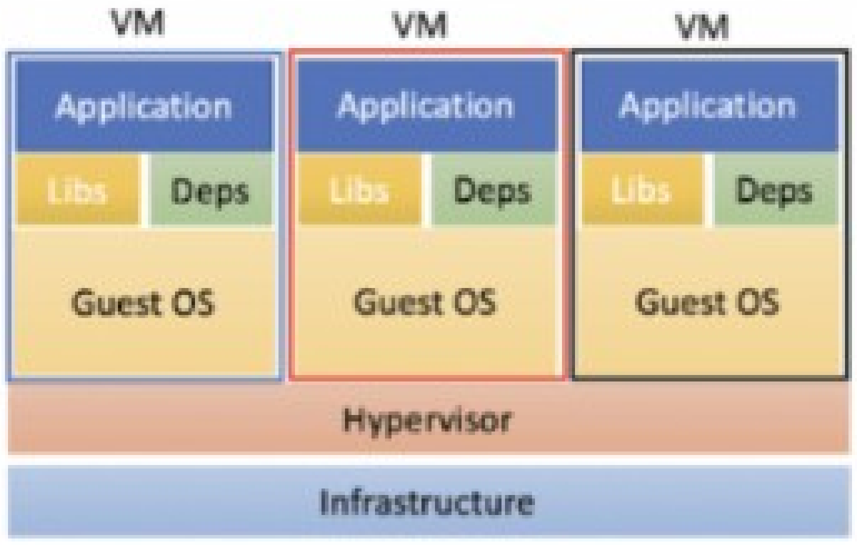
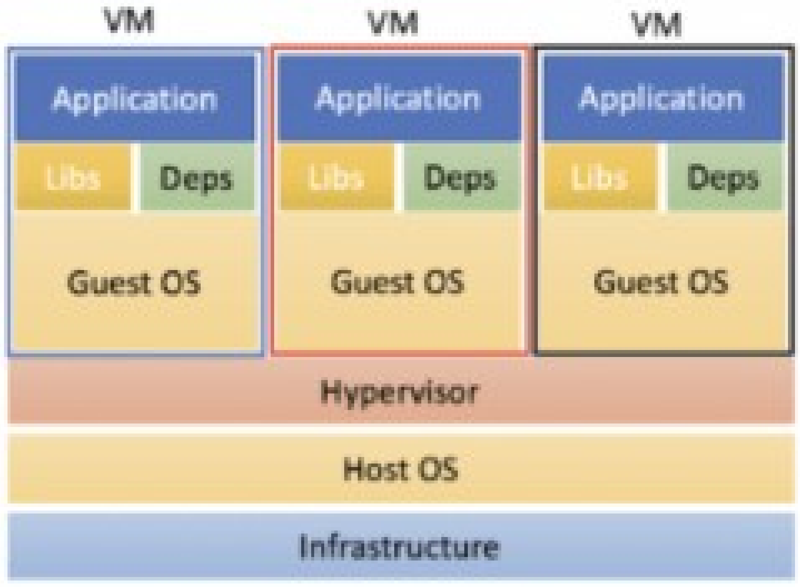
Computing Environments – Virtualization



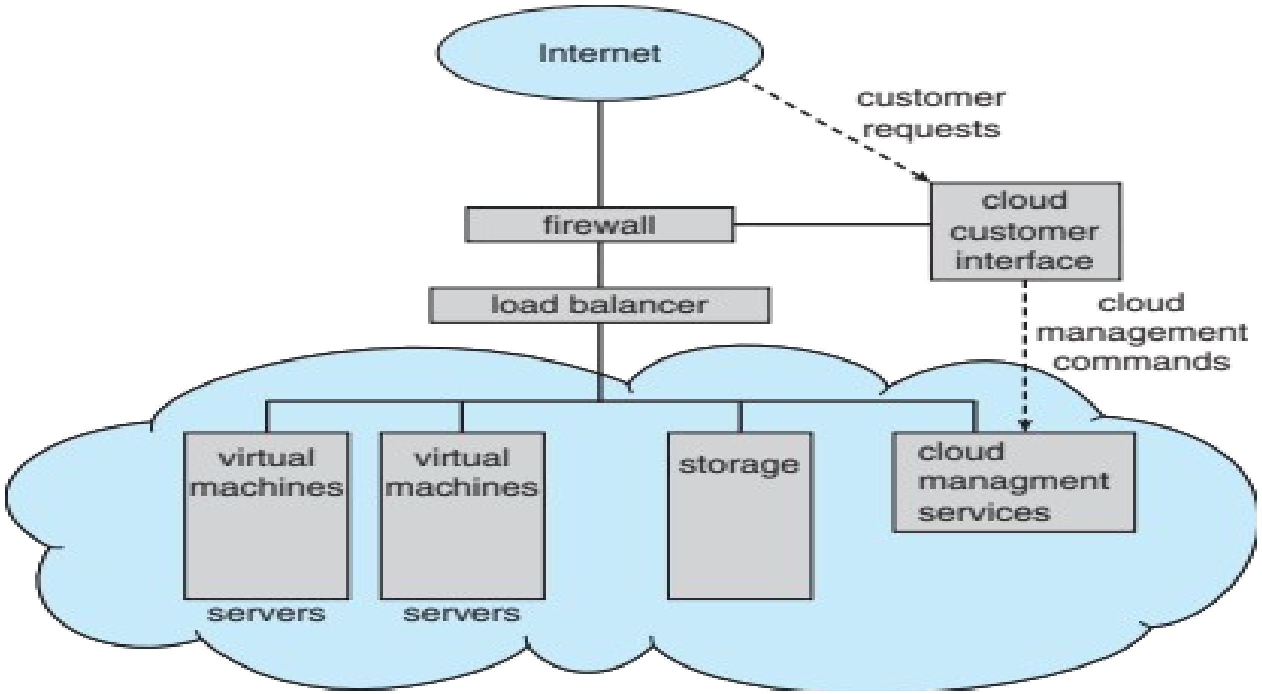
Type-1



Type-2

Computing Environments – Cloud Computing

- Cloud computing environments composed of traditional OSes, plus VMMs, plus cloud management tools
  - Internet connectivity requires security like firewalls
  - Load balancers spread



Computing Environments – Cloud Computing

- Delivers computing, storage, even apps as a service across a network
- Logical extension of virtualization because it uses virtualization as the base for it functionality.
  - Amazon **EC2** has thousands of servers, millions of virtual machines, petabytes of storage available across the Internet, pay based on usage
- Many types
  - **Public cloud** – available via Internet to anyone willing to

Computing Environments – Cloud Computing

- Software as a Service (**SaaS**) – one or more applications available via the Internet (i.e., word processor)
- Platform as a Service (**PaaS**) – software stack ready for application use via the Internet (i.e., a database server)
- Infrastructure as a Service (**IaaS**) – servers or storage available over Internet (i.e., storage available for backup use)

Computing Environments – Real-Time Embedded Systems

- Real-time embedded systems most prevalent form of computers
  - Vary considerable, special purpose, limited purpose OS, **real-time OS**
  - Use expanding
- Many other special computing environments as well
  - Some have OSes, some perform tasks without an OS
- Real-time OS has well-defined fixed time constraints
  - Processing ***must*** be done within constraint
  - Correct operation only if constraints met

THANK YOU

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